

Research Methods — Module 04: Cognition — Study Questions

Recall and Comprehension

1. Why is behavioral task design critical for testing Active Inference? What is the dissociation principle?
2. How do you manipulate precision in a behavioral task? Give two examples.
3. How do you manipulate volatility? What behavioral measures reflect volatility estimation?
4. Why are eye movements considered direct evidence of active inference?
5. What does fixation duration at a particular location reveal about uncertainty?

Analysis and Application

1. How does scanpath analysis reveal the brain's hypothesis-testing strategy?
2. What does Active Inference predict about visual search termination? How does this differ from signal detection theory?
3. How does pupil dilation relate to prediction error (surprise)?
4. What is the relationship between tonic pupil size and decision uncertainty?
5. How does the locus coeruleus-norepinephrine system connect to precision modulation?
6. How does probabilistic reversal learning dissociate learning rate from volatility estimation?
7. What is the "sampling problem"? How does Active Inference provide a stopping rule?

Research Design

1. What determines planning depth in Active Inference? How might you measure it experimentally?
2. What is a MouseLab paradigm? How does it reveal information sampling strategies?
3. What control conditions are necessary for a volatility manipulation? Why is a stable control block insufficient on its own?

4. How does computational phenotyping address individual differences in behavioral tasks?
5. What confounds threaten the interpretation of pupil dilation as "surprise"? Name at least three (luminance, arousal, cognitive load).
6. How do practice effects affect computational parameter estimates? When should they be modeled explicitly?

Critical Evaluation

1. Design a behavioral task dissociating epistemic value from pragmatic value. Specify: (a) the state space, (b) the manipulation, (c) the dependent variable, and (d) the predicted dissociation.
2. Critically evaluate: Can behavioral data alone distinguish Active Inference from reinforcement learning accounts? What additional evidence would you need?
3. A study reports that patients with schizophrenia show reduced volatility learning rates compared to controls. What alternative interpretations exist? How would you rule them out?
4. How does the "rich get richer" problem (accumulating evidence bias in drift-diffusion models) compare to precision accumulation in Active Inference models of decision-making?
5. You find that a model with state-dependent precision fits behavioral data better than a model with fixed precision. What does this say about the cognitive process — and what doesn't it tell you?
6. Design a minimal behavioral experiment (≤ 3 conditions) to test whether exploration in a two-armed bandit is driven by epistemic value (uncertainty reduction) vs. optimistic bias. Specify the critical comparison.
7. A reviewer argues that your Active Inference model of a cognitive task has too many free parameters compared to a simpler drift-diffusion model. How do you respond? What role does Bayesian model comparison play?